

CMPUT 429/530 Computer Systems and Architecture

Course Outline

General Information

Term: Winter, 2025, Lecture B1
Date and Time: MWF at 11-12am
Location: Tory Building (T B-95)
Number of credits: 3 credits

See the contact information for details about your instructor and teaching assistants.

Overview

Official Course Description: *An investigation of computer system design concepts including requirements, specifications, implementation, and modification. Instruction sets, arithmetic/logic unit design, bus structures, I/O structures, control organization, and implementation. Discussion and use of hardware description languages.*

The course focuses on the most important concepts employed in modern multi-CPU computers. It also discusses the implications of hardware structures in the writing of programs and on the generation of code by compilers.

Objectives

To give you an understanding of the contemporary solutions and trends in computer systems design and their role in facilitating high-performance computing. In addition to becoming familiar with those solutions and their objectives, you will also become familiar with key issues that have to be addressed by compilers to generate efficient code for high-performance multi-CPU computers.

Pre-requisites

CMPUT 201, 229 or EE 380. Credit may be obtained in only one of CMPUT 429 or CMPE 382.

Course Topics

- Review of Performance Evaluation
- Pipelining, Caches, Virtual Memory, and Paging
- Superscalar Processors
- Branch Prediction, Instruction Fetching, and Register Renaming
- Instruction Scheduling, Memory Accesses
- Memory Hierarchy
- Multiprocessors
- Multithreading and Chip Multiprocessing

Course Work and Evaluation

Course Work	Weight
Assignments	23%
Midterm	32%
Homework and Quizzes	6%
Final Exam	39%

Assignment of Final Letter Grades

To assign final letter grades the numerical grades obtained from the course work listed above are sorted in decreasing order. An initial distribution of letter grades following the historical distribution of letter grades from previous editions of CMPUT 429 taught by the same professor is made. After this initial distribution, the cut points for letter grades are adjusted to reflect natural gaps found in the distribution of numerical grades for the current edition of the course. This final adjustment also considers the level of difficulty of the midterm and final in the current term based on the overall performance of the class in these exams.

Course Materials

Required Textbook:

Baer, Jean-Loup, *Microprocessor Architecture: From Simple Pipelines to Chip Multiprocessors*, Cambridge University Press, First Edition, 2010.

Academic Integrity

The University of Alberta is committed to the highest standards of academic integrity and honesty. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect. Students are particularly urged to familiarize themselves with the provisions of the Code of Student Behaviour (online at www.ualberta.ca/secretariat/appeals.htm) and avoid any behaviour which could potentially result in suspicions of cheating, plagiarism, misrepresentation of facts, and/or participation in an offense. Academic dishonesty is a serious offense and can result in suspension or expulsion from the University. (GFC 29 SEP 2003)

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